OVIDIUS UNIVERSITY OF CONSTANȚA

Faculty of Mathematics and Computer Science · Computer Science

BACHELOR'S THESIS

Information System for Network Detection Intrusion

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Project purpose

Real-time detection

Each packet inspected as it arrives. No PCAP files analysed offline.

Hybrid signatures + ML

Two complementary engines. When both agree, the alert gets the hybrid label.

Built-in forensics

Per alert: a snapshot of the last 50 packets, downloadable from the UI.

Easy demo

One-click launcher: auto-install, FastAPI + Vite start, browser opens.

Project idea

Neither classical paradigm solves the problem on its own.

SIGNATURES

Precise but limited

Catch exactly what they know (CVEs, Snort signatures). Cannot see new or modified attacks.

MACHINE LEARNING

Flexible but noisy

Detects unknown anomalies. But generates false alarms — see Sommer & Paxson, 2010.

HYBRID

Explicit cooperation

When both engines flag the same packet → the hybrid category (maximum confidence).

Plus a built-in forensics layer — every alert keeps the context of the packets that triggered it.

Technologies used

Backend

Python 3.12	Main language, async event loop
FastAPI + Uvicorn	REST + WebSocket, auto OpenAPI
SQLAlchemy 2 async	ORM + aiosqlite (SQLite)
scikit-learn	Random Forest, Isolation Forest
Scapy	Real-time packet capture (libpcap / Npcap)
bcrypt + JWT	Authentication and password hashing

Frontend

React 18 + Vite	SPA, hot reload in dev
TailwindCSS	Deep Navy + Cyan theme
Recharts	Live charts (packet rate, severity)
lucide-react	Consistent iconography
WebSocket client	Streaming for alerts + traffic
date-fns	Timestamp formatting

Packet flow through the system



Capture

Two interchangeable sources: LiveSniffer (Scapy) or TrafficSimulator (in-memory). Same API.

Processing

Dispatcher: metrics, 200-packet ring buffer, traffic_bus publishing, DB sampling.

Detection

Signature engine (10 YAML rules) + ML scoring. Three categories: signature, ml, hybrid.

Persistence

Alerts in SQLite + JSONL per alert on disk. Broadcast on alert_bus → WebSocket clients.

End-to-end latency (packet → UI notification) under 250 ms on a laptop.

The modules that make up the application

Sniffer / Simulator

Real packet capture (Scapy) or synthetic generation for demos.

Dispatcher

The central point. Coordinates metrics, engines, persistence, broadcast.

Signature Engine

Editable rules (protocol, IP, port, content, regex, threshold).

ML Detector

Random Forest or Isolation Forest on 6 features per packet.

Forensics

200-packet ring buffer; per alert, JSONL written to disk.

API + WebSocket

JWT-protected REST (/api/...) + /ws/alerts + /ws/traffic streaming.

The detection pipeline

Per-packet processing

1. Dispatcher.handle(pkt) is called for every captured packet.
2. Increment counters + update PPS (1 s sliding window).
3. Packet enters the 200-slot ring buffer (source of forensic snapshots).
4. Publish on traffic_bus → WebSocket → live dashboard chart.
5. 1/5 sampling persisted as TrafficEvent (for history).
6. Signature engine iterates the compiled rules (10 defaults).
7. ML detector (if active) extracts features and scores the packet.
8. Merge: signature-only, ml-only or hybrid → alerts created.

Alert categories

signature

Only the signature engine fired. Precise, auditable.

ml

Only the ML model scored ≥ 0.8. Suspicious packet, no known rule.

hybrid

Both engines agreed. Maximum confidence, top triage priority.

Two complementary algorithms

Random Forest

supervised

LEARNING	Labelled data (attack vs benign)
OUTPUT	Label + calibrated probability
STRENGTH	Interpretable feature importance
WEAKNESS	Cannot detect unknown attacks

Isolation Forest

unsupervised

LEARNING	Normal traffic only — no attack labels needed
OUTPUT	Anomaly score (higher = more suspicious)
STRENGTH	Detects unknown outliers
WEAKNESS	Higher false-positive rate

Feature vector (6 dimensions per packet):

`src_port` `dst_port` `length` `proto_id` `flag_mask` `inter_arrival_ms`

Per-alert capture

How it works

1. Ring buffer

The dispatcher keeps a FIFO of the last 200 captured packets.

2. On every alert

The last 50 packets are extracted from the buffer and serialised as JSON Lines.

3. Association

The file is linked to the alert through a foreign key in the database.

4. Download

Authenticated endpoint /api/forensics/alert/{id}/download via JWT.

5. Replay in the UI

The Alert Detail page shows the packet timeline and payload preview.

alert_12_1716387210.jsonl

```
{"ts": "...:32.481Z", "protocol": "TCP",
 "src_ip": "45.77.12.88", "src_port": 51232,
 "dst_ip": "10.0.0.10", "dst_port": 22,
 "length": 60, "flags": "S",
 "payload_preview": ""}
{"ts": "...:32.491Z", "protocol": "TCP",
 "src_ip": "45.77.12.88", "src_port": 51233,
 "dst_ip": "10.0.0.10", "dst_port": 80,
 "length": 60, "flags": "S",
 "payload_preview": ""}
...
// 50 packets total
```

Dashboard and Attack Console

DASHBOARD

Monitoring host (simulator)

ovidius-nids-sensor · 10.0.0.10

Alerts (24h)	Open	Packets	PPS
47	12	12.4k	31.2

LIVE PACKET RATE



ATTACK CONSOLE

Nmap scan
37 packets · 30 alerts

SSH brute-force
17 packets · 12 alerts

ICMP flood
64 packets · 40 alerts

SQL injection
1 packet · 1 alert

Command inject.
1 packet · 1 alert

Reverse shell
1 packet · 1 alert

From attack to triage

1 · Attack triggered

The user clicks "Nmap port scan" in the Attack Console. The simulator generates 30 TCP SYN packets from 185.220.101.5 toward 10.0.0.10.

2 · Capture

The dispatcher receives each packet, updates the ring buffer and publishes on traffic_bus for the live chart.

3 · Signature detection

After the 8th packet, the "Nmap TCP SYN scan" rule crosses the threshold (8 SYNs in 5 s) — the alert is created.

4 · Persistence + forensics

The alert is saved in SQLite. A JSONL ring-tail with the last 50 packets is written to disk and linked to the alert.

5 · Real-time notification

alert_bus publishes on WebSocket. The Alerts page receives the alert instantly via /ws/alerts.

6 · Triage

The analyst opens Alert Detail, sees the packet timeline and downloads the JSONL capture for investigation.

Similar applications on the market

Feature	Snort	Suricata	Zeek	Wazuh	Splunk ES	Ovidius NIDS
Signature engine	✓	✓	~	~	~	✓
Machine Learning engine	✗	✗	✗	~	✓	✓
Hybrid category (signature + ML)	–	–	–	✗	✗	✓
Built-in analyst UI	✗	✗	✗	✓	✓	✓
Per-alert forensic capture	✗	✗	✗	✗	✗	✓
Real-time streaming in UI	✗	✗	✗	~	✓	✓
Rules editable from UI (hot reload)	✗	✗	✗	✓	✓	✓
One-click deploy (Win/Mac/Linux)	✗	✗	✗	✗	✗	✓
Auto-hosted / on-prem	✓	✓	✓	✓	~	✓
Open source · free	✓	✓	✓	✓	✗	✓

All in one binary

Detection + ML + forensics + UI in a single application. The others require integration with separate tools.

Forensics out of the box

The only one in the list that automatically saves the packets around every alert.

Explicit hybrid

The hybrid category when both engines agree — does not exist in Snort / Suricata / Zeek.

Five-minute demo

One-click launcher, no external tooling, ideal for academic evaluation or showcase.

✓ full · ~ partial · – n/a · ✗ none

What was achieved

End-to-end functional

Capture, hybrid detection, forensics, authentication, live dashboard — all operational.

Real-time, low latency

< 1 ms per packet at the signature engine, < 250 ms total latency from packet to UI.

ML trainable from UI

Random Forest and Isolation Forest, with metrics and feature importance visualisation.

Forensics on disk

Per alert: a JSONL of the last 50 packets, authenticated download from the UI.

One-click deployment

run.bat / run.sh / run.py — automatic install and start (Win/Mac/Linux).

Extensible code

Rules editable from the UI without restart. Decoupled modules. Auto-generated OpenAPI.

What is left until the defense

PRIORITY

Per-flow features

Switch from per-packet features to per-flow 5-tuple (duration, packet count, total bytes). Improves precision on multi-packet attacks.

PRIORITY

Evaluation on CICIDS2017

Reproduce the training and testing pipeline on CICIDS2017 + NSL-KDD for comparability with the literature.

PRIORITY

Alembic migrations

Replace create_all() with a versioned migration system for schema evolution without data loss.

STRETCH

Alert correlation

Group consecutive alerts from the same source into a single "incident".

STRETCH

Snort rule import

Translate a subset of the Snort ecosystem into the internal JSON specification.

STRETCH

LLM-assisted triage

Chat panel on Alert Detail that summarises captured packets in natural language.



Thank you for your attention!

Bora Cristian-Vasile · Ovidius University of Constanța · 2026